

Project Proposal: Arduino-Based Robotic Face with LED Expressions

Project Type: Electronics, Programming, and Robotics

Submitted To: Electronics and Robotics Lab

Purpose: Project Approval and Equipment Request

1. Project Overview

I am proposing the development of a small robotic face capable of displaying different facial expressions and performing basic head movements. The project will use an Arduino Nano, an Adafruit QT Py SAMD21 development board, a 5×5 NeoPixel LED matrix, and a two-axis servo pan-tilt kit.

The goal is to explore how microcontrollers, LED displays, and servo motors can work together to create an interactive robotic system. The finished project will demonstrate basic robotics concepts, communication between microcontrollers, and hardware integration.

2. Project Objectives

The primary objectives of this project are to:

- Program a 5×5 LED matrix to display basic facial expressions, including happy, sad, neutral, and surprised.
- Use two servo motors to control horizontal and vertical head movement.
- Develop a simple control interface that allows a user to change expressions and control movement.
- Establish communication between the Arduino Nano and QT Py development board.
- Practice circuit assembly, troubleshooting, programming, and system testing.

3. Equipment and Materials

Component	Qt	Purpose
Arduino Nano	1	Controls servo movement
Adafruit QT Py SAMD21	1	Controls the LED matrix
Adafruit 5x5 NeoPixel Grid BFF	1	Displays facial expressions
Pan-Tilt Kit with SG90 Servos	1	Provides horizontal and vertical movement
Breadboard and Jumper Wires	1 set	Circuit connections
Regulated 5V Power Supply	1	Provides sufficient power for the servos
Logic Level Shifter	1	Supports safe communication between the microcontrollers
USB Data Cables	2	Programming and initial testing
Mounting Hardware	As needed	Secures the LED display to the pan-tilt assembly

4. Proposed Build Process

The project will be completed in four phases.

Phase 1: LED Display Programming

Connect the NeoPixel matrix to the QT Py development board and program it using the Arduino IDE. Initial testing will focus on displaying solid colors, followed by simple facial expressions created using individual LEDs.

Phase 2: Servo Motor Assembly

Assemble the pan-tilt mechanism and connect both servo motors to the Arduino Nano. Develop a basic program to control horizontal and vertical movement while establishing safe movement limits.

Phase 3: System Integration

Mount the LED matrix to the pan-tilt mechanism and connect the two microcontrollers using a suitable communication method. Initially, both boards may be controlled independently through USB to simplify testing. A shared command system will then be developed to coordinate expressions with movement.

Phase 4: Testing and Demonstration

Test the completed system for reliable movement, correct LED patterns, and consistent responses to user commands. The final demonstration will show the robotic face changing expressions while moving in different directions.

5. Safety and Technical Considerations

The project will operate using low-voltage electronics. Servo motors will receive power from an appropriate regulated supply rather than directly from a microcontroller's GPIO pins.

Because the Arduino Nano and QT Py use different logic voltage levels, appropriate voltage-level conversion will be used when connecting their communication pins.

All wiring and power connections will be reviewed before testing. The pan-tilt mechanism will also be programmed with movement limits to reduce the possibility of mechanical damage.

6. Expected Results

Upon completion, the project should demonstrate a functional robotic face capable of:

- Displaying at least four recognizable facial expressions.
- Moving horizontally and vertically through programmed servo commands.
- Responding to user input through a basic computer interface.
- Coordinating LED expressions with physical movement.

The completed system will serve as a foundation for future improvements, such as sensors, automated movement routines, or more advanced animations.